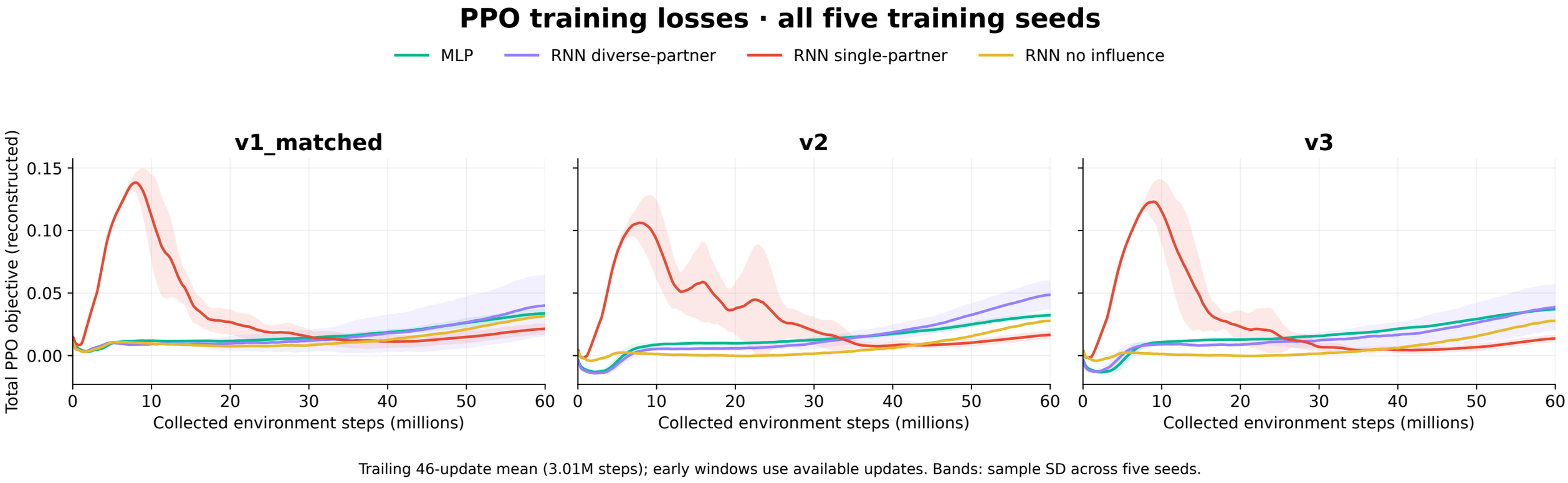

PPO training losses · all five training seeds
MLP
RNN diverse-partner
RNN single-partner
RNN no influence
v1_matched
v2
v3
Total PPO objective (reconstructed)
0.15
0.10
0.05
0.00
0
10
20
30
40
50
60
Collected environment steps (millions)
Trailing 46-update mean (3.01M steps); early windows use available updates. Bands: sample SD across five seeds.